Besov Spaces over Good Grids

Current Lean 4 Formalization

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Abstract

This document describes the current Lean 4 formalization in the `BesovSpacesGoodGrid` repository. The project develops a reusable weak-grid framework for atomic Besov-ish spaces, proves scale inclusions, representation-limit and compactness theorems, proves completeness for the coefficient-cost norm, and formalizes a substantial weak-grid transmutation pipeline. It also contains a good-grid specialization based on Souza atoms. The principal public results are summarized in Section 10.

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1 Introduction

1.1 Purpose of the Project

The project formalizes part of the atomic decomposition theory from

Daniel Smania, *Besov-ish spaces through atomic decomposition*, Analysis & PDE 15 (2022), no. 1, 123–174.

The emphasis is on a Lean library that can be reused in later developments: the basic theory is first stated for weak grids and abstract atom families, and good-grid/Souza spaces are then obtained as a specialization.

1.2 Current Pipeline

The root Lean file `BesovSpacesGoodGrid.lean` currently imports the following pipeline:

1. define quantitative good grids and general weak grids;
2. package weak-grid cells and local Banach spaces;
3. define atom families with support, convexity, phase invariance, and atom-size estimates;
4. build levelwise atomic blocks and infinite L^p representations;
5. define the Besov-ish space and the coefficient-cost gauge `Norm_Costpq`;
6. prove scale constructions and a smoothness-scale inclusion;
7. prove representation-limit theorems, compactness statements, and completeness of the Besov-ish space for the cost norm;
8. prove transmutation claims moving atomic representations from one weak grid to another.

The repository also contains good-grid Souza-space results and auxiliary block-sum notation. Those files are documented here because they are part of the current repository state, even though not all of them are imported by the root module yet.

2 Grid Structures

2.1 Good Grids

Definition 2.1 (Good grid)**.** The Lean structure

`GoodGridSpace.GoodGrid`

extends the grid structure from the `UnbalancedHaarWavelet` dependency. It adds constants $\lambda_1, \lambda_2 : \mathbb{R}$ with

$$0 < \lambda_1 \leq \lambda_2 < 1$$

and lower and upper measure-ratio estimates for a child cell s inside a parent cell t:

$$\lambda_1 \, \mu(t) \leq \mu(s), \qquad \mu(s) \leq \lambda_2 \, \mu(t).$$

Definition 2.2 (Good-grid space)**.** The structure

`GoodGridSpace.GoodGridSpace`

is the ambient object used by good-grid files. It consists of a measurable space with a fixed good grid. The associated measure is exposed as `GoodGridSpace.measure`.

2.2 Weak Grids

Definition 2.3 (Weak grid)**.** The Lean structure

`WeakGridSpace.WeakGrid`

is a measure together with a sequence of finite cell families $\mathcal{P}^k$. Each cell is measurable and has positive measure; the whole measure is finite; at least one level is nonempty; and there is a uniform same-level overlap bound `Cmult1`.

The term "partition" appears in field names because of the mathematical source, but a weak grid does not require disjointness or covering. The important primitive is the overlap neighborhood

$$\Omega_Q^k = \{P \in \mathcal{P}^k : P \cap Q \neq \varnothing\},$$

implemented as `overlapFinset`.

Lemma 2.1 (Overlap double counting)**.** The theorem

`WeakGridSpace.WeakGrid.overlap_double_sum_le`

states that if f is nonnegative on the level-k cells, then summing f over all overlap neighborhoods costs at most the multiplicity constant:

$$\sum_{Q \in \mathcal{P}^k} \sum_{P \in \Omega_Q^k} f(P) \leq C_{\text{mult1}} \sum_{P \in \mathcal{P}^k} f(P).$$

3 Atoms

3.1 Cells and Local Spaces

Definition 3.1 (Weak-grid cell)**.** The structure

`WeakGridSpace.WeakGridCell`

bundles a level k, a set Q, and the proof that $Q \in \mathcal{P}^k$. The theorem `measurable` records that every such cell is measurable.

Definition 3.2 (Local Banach space)**.** The structure

`WeakGridSpace.LocalBanachSpace`

packages a complex normed complete vector space together with an injective linear realization map into functions $\alpha \to \mathbb{C}$. This lets different cells carry different local model spaces.

3.2 Atom Families

Definition 3.3 (Atom family)**.** The Lean structure

`WeakGridSpace.AtomFamily`

is the central abstraction for atomic decompositions. For parameters s, p, u, it records:

(i) the Holder conjugate u';

(ii) assumptions $s > 0$, $1 \le p < \infty$, and $1 \le u \le \infty$;

(iii) a local Banach space $B(Q)$ for every weak-grid cell Q;

(iv) a nonempty atom set $A(Q) \subset B(Q)$;

(v) membership of local functions in L^{pu};

(vi) support inside Q;

(vii) convexity and phase invariance of the atom set;

(viii) the atom-size estimate
$$\|a\|_{L^{pu}} \le \mu(Q)^{s-1/(u'p)}.$$

The namespace `AtomFamily` exposes convenience definitions and lemmas: `toFunction`, `IsAtom`, `AtomsOn`, `allAtoms`, `atom_memLp`, `local_memLp_p`, `atom_support`, and `atom_norm_bound`. It also records optional compactness and finite-dimensional hypotheses: `StronglyCompactAtoms`, `WeaklySequentiallyCompactAtoms`, and `FiniteDimensionalAtoms`.

4 Besov-ish Spaces

4.1 Level Blocks

Definition 4.1 (Level block)**.** For an atom family A, the structure

```
WeakGridSpace.LevelBlock A k
```

chooses one coefficient and one atom for each cell of the level k. Its value in L^p is the finite sum over that level:

$$\sum_{Q \in \mathcal{P}^k} c_Q a_Q.$$

Lean names this value `LevelBlock.toLp`.

The file proves that zero and scalar multiples of level blocks behave as expected. It also proves basic closure properties of the set `LevelBlockSet A k`.

4.2 Representations and Costs

Definition 4.2 (Atomic representation)**.** The structure

```
WeakGridSpace.LpGridRepresentation A g
```

is a sequence of level blocks whose infinite sum is g in L^p:

$$\mathrm{HasSum}\,(k \mapsto \mathrm{toLp}(B_k))\,(g).$$

Definition 4.3 (Coefficient cost)**.** For a representation R, Lean defines `levelCoeffPower`, `lpCost`, `pqCost`, and `pqCostENNReal`. The finite-cost predicate is

```
LpGridRepresentation.FinitePQCost.
```

It is the formal version of the usual (p, q)-coefficient summability condition, with the endpoint $q = \infty$ represented by a supremum bound.

4.3 Besov-ish Space and Cost Gauge

Definition 4.4 (Besov-ish membership)**.** The predicates

```
MemBesovish,     MemBesovishCoeffCost
```

say that an L^p function admits an atomic representation, respectively an atomic representation with finite (p, q)-coefficient cost.

Definition 4.5 (Besov-ish space)**.** The Lean definition

```
WeakGridSpace.BesovishSpace A q
```

is the complex submodule of L^p consisting of elements satisfying `MemBesovishCoeffCost A q`.

Definition 4.6 (Coefficient-cost gauge)**.** For $g \in$ `BesovishSpace A q`, the gauge

```
BesovishSpace.Norm_Costpq A q g
```

is the infimum of the pqCost over all finite-cost representations of g.

Theorem 4.1 (Normed-space summary)**.** The theorem

`BesovishSpace.normedSpace_and_lp_embedding`

packages the main structural facts about `Norm_Costpq`: nonnegativity, triangle inequality, homogeneity, a named cost-normed group/space structure, and an explicit L^t embedding estimate for exponents

$$p \le t \le pu.$$

5 Scale Inclusions

Definition 5.1 (Smoothness scaling)**.** The file `WeakGridScalesBesovSpaces.lean` defines

`smoothnessScaleFactor`, `smoothnessScaleAtoms`, `smoothnessScaleAtomFamily`.

These rescale atoms cell by cell by the deterministic factor

$$\mu(Q)^{\tilde{s}-s}.$$

Definition 5.2 (Smoothness-integrability scaling)**.** When $u = \infty$, the file also defines the two-parameter scaling

`smoothnessIntegrabilityScaleAtomFamily`.

The scale factor is

$$\mu(Q)^{\tilde{s}-s+1/p-1/\tilde{p}}.$$

Theorem 5.1 (Smoothness-scale inclusion)**.** The theorem

`smoothnessScaleBesovishSpace_subset`

shows that, under the deterministic summability hypothesis `scaleCoefficientFinite` and the monotonicity assumption $s \leq \tilde{s}$, the Besov-ish space for the smoothness-scaled atom family is contained in the original Besov-ish space.

Theorem 5.2 (Linear inclusion and cost bound)**.** The definition

`smoothnessScaleBesovishSpaceInclusion`

packages the preceding set inclusion as a linear map. The theorem

`smoothnessScaleBesovishSpaceInclusion_Norm_Costpq_le`

gives the corresponding explicit bound for the cost gauge.

6 Representation Limits and Completeness

6.1 Analytic Grid Hypothesis

Definition 6.1 (Assumption G2)**.** The predicate

`WeakGridSpace.AssumptionG2 G s p u q`

combines two requirements:

(i) the coefficient-weight series needed for the $t = p$ embedding is finite;

(ii) the maximal measure of cells at level k tends to zero as $k \to \infty$.

This hypothesis is used to control tails of atomic representations uniformly.

6.2 Limit Theorems

Theorem 6.1 (Representation limit)**.** The theorem

`representation_limit`

formalizes the weak L^p version of Proposition `compa2` from the development. If representations have uniformly bounded coefficient cost, their coefficients converge cellwise, and their atoms converge weakly, then the limiting blocks define a Besov-ish function with the same cost bound and the represented functions converge weakly.

Theorem 6.2 (Strong representation limit)**.** The theorem

`representation_limit_strong`

is the strong L^p version: strong convergence of the atoms gives strong convergence of the represented functions.

The existence theorems `representation_limit_strong_existence` and `representation_limit_w` start from a bare limit block sequence, prove it is summable, and package the resulting limit as a genuine representation.

6.3 Compactness and Completeness

Theorem 6.3 (Formal block summability)**.** The theorem

`formalBlockSeq_summable`

says that under `AssumptionG2`, any formal block sequence with finite abstract (p, q)-cost is summable in L^p.

Theorem 6.4 (Closed cost balls)**.** The theorem

`closed_Norm_Costpq_ball_strongly_seqCompact`

proves sequential compactness of closed coefficient-cost balls in the ambient strong L^p topology, assuming the compactness hypothesis `AssumptionA5`.

Theorem 6.5 (Completeness)**.** The theorem

`besovishSpace_costNorm_completeSpace`

states that `BesovishSpace A q`, endowed with the local `costNormedAddCommGroup` structure associated to `Norm_Costpq`, is complete.

7 Weak-Grid Transmutation

7.1 Setup

The file `WeakGridTransmutation.lean` formalizes the transfer of an atomic representation from a source weak grid G to a target weak grid W. The source coefficients are measured by `CoeffPLevel`, `CoeffPQCost`, and `CoeffFinitePQCost`.

Definition 7.1 (Almost linear level selector)**.** The predicate

```
AlmostLinearSequence k
```

says that the target level $k(i)$ attached to a source level i stays between two affine functions with the same positive slope:

$$ri + A \leq k(i) \leq ri + B.$$

It is used to prove finiteness of certain level interactions and to reduce coefficient estimates to convolution bounds.

Definition 7.2 (Representation hypothesis)**.** The predicate

```
RepresentationWsubGandALS
```

records the structural hypothesis that every source atom has a target-grid atomic representation with localization, decay, and almost-linear level control.

7.2 Transmutation Objects

For each truncation level N, the file defines:

- `PartialSumLevels`: the source partial expansion;
- `TransmutationCoeff`: the target coefficient $m_{P,N}$;
- `TransmutationAtom` and `TransmutationAtomLocal`: the normalized target atoms $d_{P,N}$;
- `TransmutationBlock`: the target level block at level j.

It also defines stable limiting objects: `TransmutationCoeffLimit`, `TransmutationAtomLimit`, and `TransmutationBlockLimit`.

7.3 Claims

Theorem 7.1 (Claim I)**.** The theorem

```
ClaimI
```

is the exact bookkeeping identity. The transmuted target representation at truncation N sums to the source partial expansion `PartialSumLevels ... N`. The theorem `ClaimI_top` is the endpoint $q = \infty$ version.

Theorem 7.2 (Claim II)**.** The theorem

```
ClaimII
```

states that each finite transmutation block sequence is a genuine Besov-ish representation of the source partial expansion and that its (p, q)-cost is bounded by the source coefficient cost times explicit geometric, decay, and convolution constants. The theorem `ClaimII_top` treats the endpoint $q = \infty$.

Theorem 7.3 (Claim III)**.** The theorem

```
ClaimIII
```

passes to the stable limiting transmutation data. It produces an L^p-limit g_∞, proves that the stable blocks form a Besov-ish representation of g_∞, and proves strong convergence of the truncated source sums to that limit. The theorem `ClaimIII_top` is the endpoint version.

8 Good-Grid Souza Spaces

The file `GoodGridBesovSpace.lean` specializes the weak-grid theory to good grids and Souza atoms.

Definition 8.1 (Induced weak grid)**.** The definition

`GoodGridSpace.toWeakGrid`

turns a good grid into a weak grid. Since good-grid partitions are disjoint, the same-level overlap multiplicity is 1. The induced weak-grid space is `GoodGridSpace.toWeakGridSpace`.

Definition 8.2 (Souza atoms)**.** The predicate

`GoodGridSpace.IsSouzaAtom`

says that an atom is supported on a good-grid cell, is constant on that cell, and has size at most

$$\mu(Q)^{s-1/p}.$$

The definitions `canonicalSouzaAtom`, `souzaLocalBanachSpace`, `souzaAtomsSet`, and `souzaAtomFamily` package these atoms as an `AtomFamily` with $u = \infty$.

Definition 8.3 (Souza Besov space)**.** The definition

`GoodGridSpace.SouzaBesovSpace`

is the Besov-ish space associated to the Souza atom family on the weak grid induced by the good grid.

Theorem 8.1 (Good-grid completeness)**.** The theorem

`souzaBesovSpace_costNorm_completeSpace`

specializes the weak-grid completeness theorem to Souza atoms on a good grid.

Theorem 8.2 (Compact balls)**.** Theorems

`souza_closedCostBallInLp_isSeqCompact`,
`souza_closedCostBallInLp_isCompact`, `souza_closedCostBallInL1_isCompact`

state compactness of closed cost balls in the ambient L^p topology and, after the finite-measure inclusion, in L^1.

Theorem 8.3 (Density)**.** Theorems

`souzaBesovSpace_dense`, `souzaBesovSpace_dense_inL1`

prove density of the Souza Besov space in the ambient L^p space and density of its image in L^1.

9 Auxiliary Files

9.1 Distributions

The file `GoodGridDistributionDefinition.lean` defines:

- `atomIndicator`: a simple-function indicator attached to a good-grid cell;
- `atomIndicators`: the set of all such indicators;
- `TestFunctions`: the real submodule spanned by the indicators;
- `Distributions`: the algebraic dual of the test-function module.

9.2 Block-Sum Notation

The file `Sums.lean` provides reusable indexing types and notation for block sums:

- `BlockIndex` for global pairs (k, Q);
- `ChildIndex` and `ParentIndex` for strict descendants and ancestors;
- `blockTsum`, `childTsum`, and `parentTsum`;
- notation such as `sum^b`, `sum^+`, and `sum^-` in the source file.

9.3 Repository Status

At this snapshot, a text search over the Lean files finds no `sorry`, no `admit`, and no project-local `axiom` declarations. The file `Basic.lean` currently contains only the placeholder definition `hello`.

10 Main Theorems

The most important public results currently include:

- `BesovishSpace.normedSpace_and_lp_embedding`: structural summary for the coefficient-cost gauge and the L^t embedding estimate.
- `smoothnessScaleBesovishSpace_subset` and `smoothnessScaleBesovishSpaceInclusion_Norm` inclusion and norm bound for smoothness-scaled Besov-ish spaces.
- `representation_limit` and `representation_limit_strong`: weak and strong representation-limit theorems.
- `representation_limit_weak_existence` and `representation_limit_strong_existence`: existence-packaged forms of the representation-limit statements.
- `closed_Norm_Costpq_ball_strongly_seqCompact`: sequential compactness of closed cost balls in the strong L^p topology.
- `besovishSpace_costNorm_completeSpace`: completeness of the Besov-ish space for `Norm_Costpq`.
- `ClaimI`, `ClaimII`, and `ClaimIII`, plus the `_top` endpoint versions: the weak-grid transmutation pipeline.
- `souzaBesovSpace_costNorm_completeSpace`, `souza_closedCostBallInLp_isCompact`, `souza_closedCostBallInL1_isCompact`, `souzaBesovSpace_dense`, and `souzaBesovSpace_dense_inL1`: good-grid/Souza consequences.

11 File Guide

- `BesovSpacesGoodGrid.lean`: main library entry point.
- `GoodGridDefinition.lean`: good-grid structures.
- `WeakGridDefinition.lean`: weak-grid structures and overlap estimates.
- `WeakGridAtomsDefinition.lean`: cells, local spaces, and atom families.
- `WeakGridBesovishSpaces.lean`: level blocks, representations, costs, Besov-ish spaces, and the cost gauge.
- `WeakGridScalesBesovSpaces.lean`: scaled atom families and inclusions.
- `WeakGridCompletenessBesovishSpaces.lean`: limit, compactness, and completeness theorems.
- `WeakGridTransmutation.lean`: transmutation objects and Claims I, II, III.
- `GoodGridBesovSpace.lean`: Souza atoms and good-grid Besov spaces.
- `GoodGridDistributionDefinition.lean`: test functions and distributions.
- `Sums.lean`: block-index and block-sum infrastructure.

12 Conclusion

The repository now contains a substantial weak-grid atomic-decomposition library: it defines Besov-ish spaces, develops their coefficient-cost norm, proves scale, compactness, completeness, and transmutation statements, and specializes the theory to Souza atoms on good grids. The most reusable API is currently the weak-grid layer; the good-grid files show how the abstract machinery is intended to specialize to concrete Besov spaces.